

EE228 – ELECTRICAL NETWORK ANALYSIS (ALL SECTIONS)

Midterm Exam-1 (SPRING 2016)

National University of Computer and Emerging Sciences, Lahore

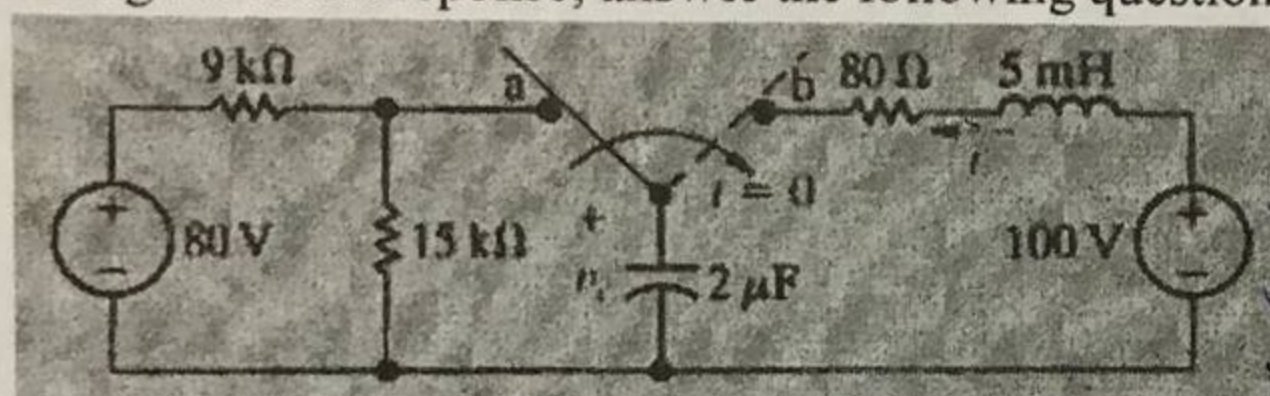
February 16, 2016

Time: 60 minutes

Note: Answer all questions with proper labeling of parts and in the specified order.

Q.No.1 [5+5+5+10=25 marks]

We are interested in finding the **total response** (transient + steady state) of the following circuit in terms of voltage across the capacitor $v_C(t)$ and the current in the circuit $i(t)$ for $t > 0$. Energy is stored in the circuit. In finding the total response, answer the following questions:



Handwritten notes: ~~Step~~ ~~Reset~~ ~~RLC~~ ~~Step Response~~ *series*

$$v_C = V_f + A_1 e^{s_1 t} + A_2 e^{s_2 t}$$

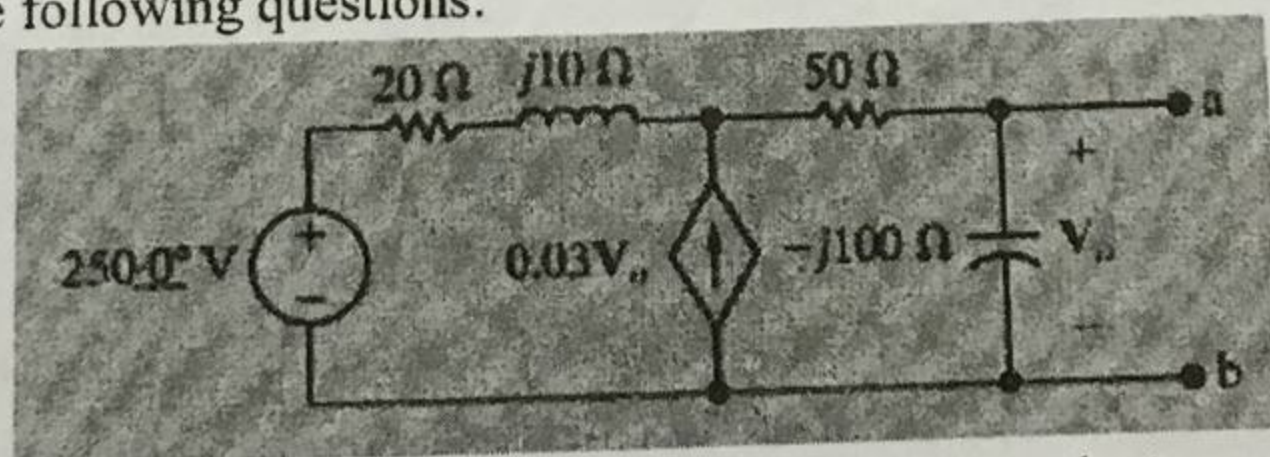
$$v_C = V_f + B_1 e^{-\alpha t} \cos \omega_d t + B_2 e^{-\alpha t} \sin \omega_d t$$

$$v_C = V_f + D_1 t e^{-\alpha t} + D_2 e^{-\alpha t}$$

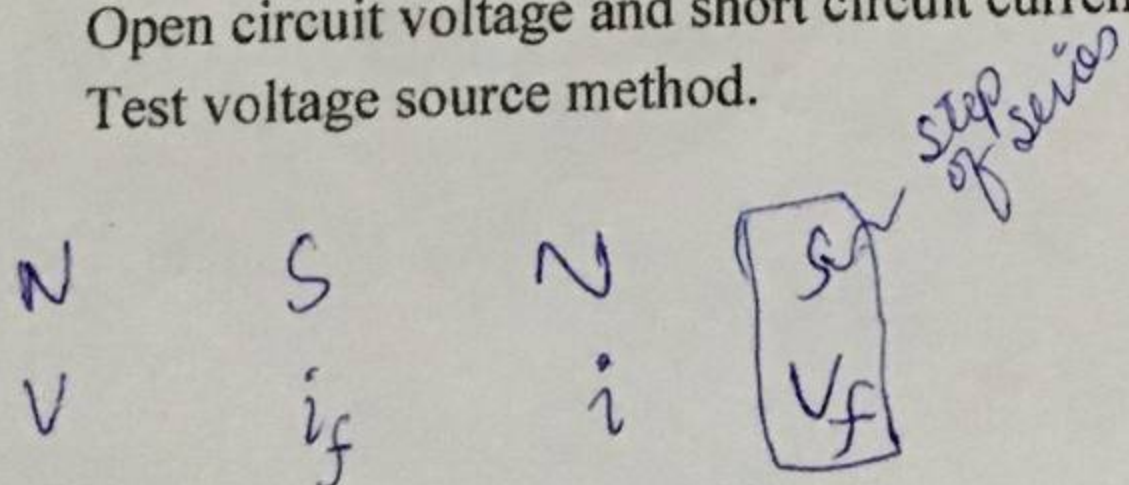
- Draw the circuit when the switch is at position 'a' for a long time before it switches to position 'b' and then find the initial conditions in the circuit shown above as follows:
 - $v_C(0+)$;
 - $i(0+)$;
 - $dv_C(0+)/dt$;
 - $di(0+)/dt$.
- Redraw the circuit at $t = 0$. The step response of the above circuit when excited by a dc voltage source is defined to be $v_C(t)$ for $t \geq 0$. Write down the differential equation (using general R, L, C, and V_s instead of their values) in terms of $v_C(t)$.
- Identify the type of response by determining the roots of the characteristic equation. Hint: $\alpha = R/2L$ and $\omega_o = 1/\sqrt{LC}$ (rad/sec).
- Determine $v_C(t)$ for $t \geq 0$ by first finding the coefficients using initial conditions:

Q.No. 2 [10+5+10=25 marks]

In the circuit shown below, we are interested in finding the **Thevenin equivalent** circuit with respect to the terminals a, b. Answer the following questions:



- Specify the domain (time/frequency) of the above circuit. Name the type of dependent source in the circuit. In addition, find the open circuit voltage V_{ab} .
- Find the short circuit current, I_N .
- Calculate the impedance Z_{ab} using:
 - Open circuit voltage and short circuit current
 - Test voltage source method.



EE228 — ELECTRICAL NETWORK ANALYSIS (ALL SECTIONS)

Midterm Exam-2 (SPRING 2016)

National University of Computer and Emerging Sciences, Lahore

April 1, 2016

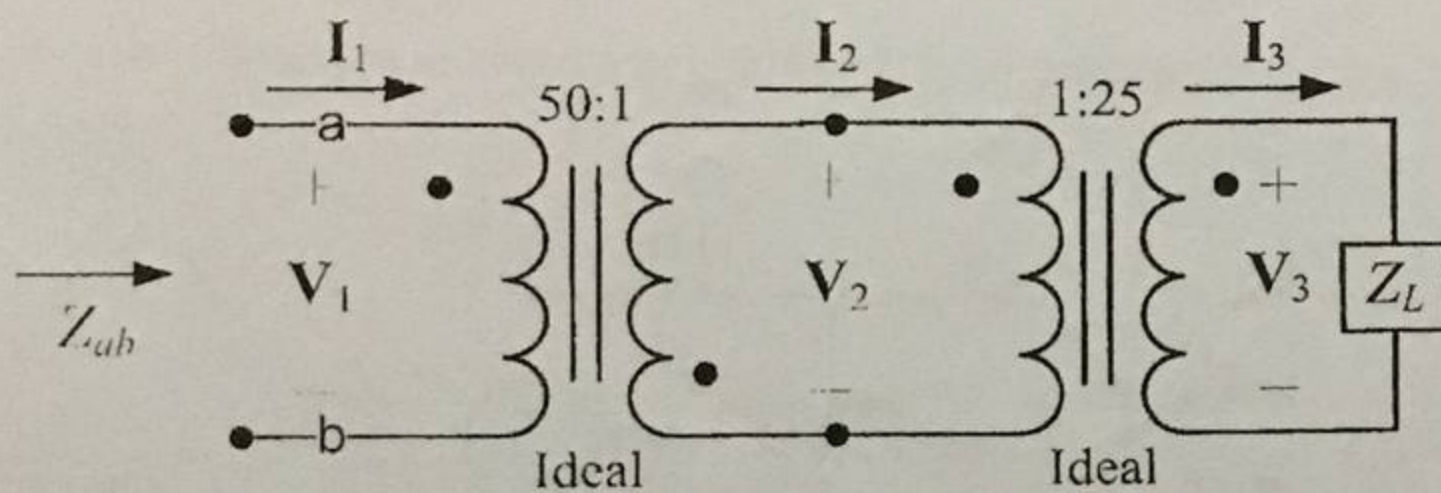
Time: 60 minutes

Note: Answer all questions.

Try to use the calculator in the intermediate calculations for precise answers in Q#2.

Q.No.1 [20 marks]

Find the impedance $Z_{ab} = V_1/I_1$ in the circuit shown below if $Z_L = V_3/I_3 = 200 + j150 \Omega$ with ideal transformers. Show all the calculations in transforming Z_L towards Z_{ab} by writing V_1/I_1 in terms of V_3/I_3 to get credit. You are not allowed to use the direct formula that relates the impedances using turns ratio except for verification.



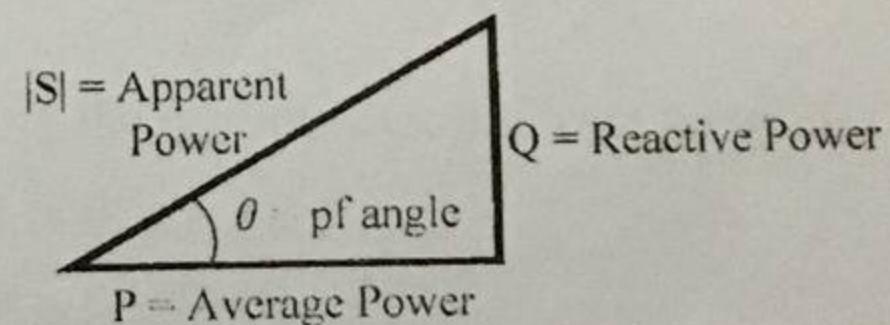
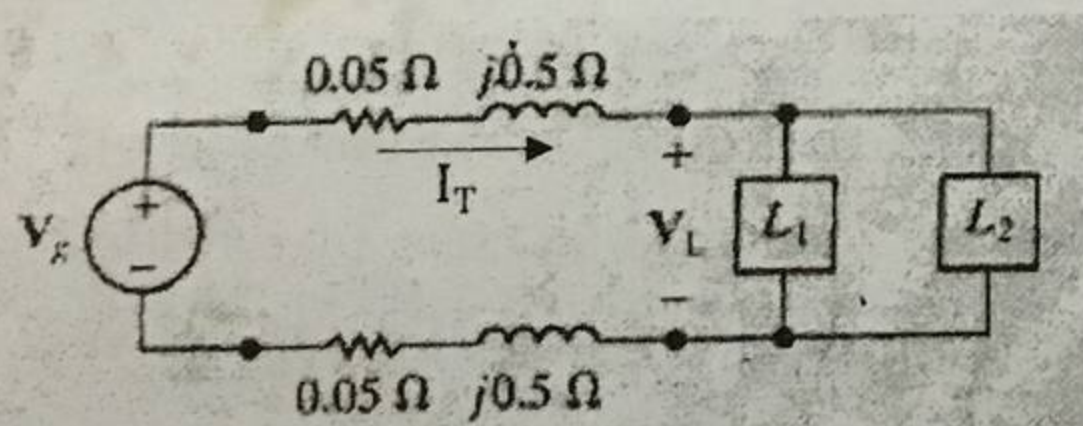
Q.No. 2 [20+5+5=30 marks]

The two loads shown in the figure below can be described as follows:

Load 1: It absorbs an average power of 60 kW and delivers 70 kVAR magnetizing reactive power.

Load 2: It has an impedance of $24 + j7 \Omega$.

The voltage at the terminals of the loads is $2500\sqrt{2} \cos 120\pi t$ V.



- Complete the power triangle for the two loads with proper labeling, subscripts (i.e. for Load 1, use P_1 , Q_1 , S_1 etc) and units. Also find the power factor for each individual load.
- Find the power factor of the combined load using the complex power 'S' of the combined load.
- Find I_T and V_g in rms.

Can you find the value of inductor of Load 2 for extra 2 marks?

Maximum Marks: 100

EE228 – Electrical Network Analysis (EE-1, 2, 3)
FINAL EXAM (SPRING 2016)

Instructors: Dr. Anjum Ali & Mr. Mohsin Yousuf

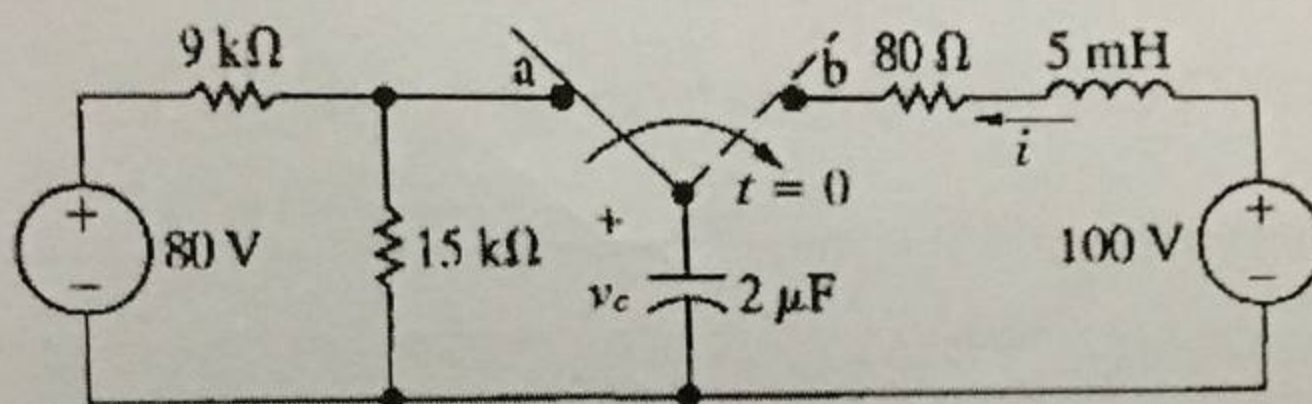
Dated: May 9, 2016

Attempt all questions with proper labeling of parts.
Total number of pages: 3

$$\alpha = \frac{1}{2RC}$$

Question # 1 [4+4+4+4+4 = 20 marks]

The switch in the circuit shown below has been in position 'a' for a long time. At time $t = 0$, it moves to position 'b'.

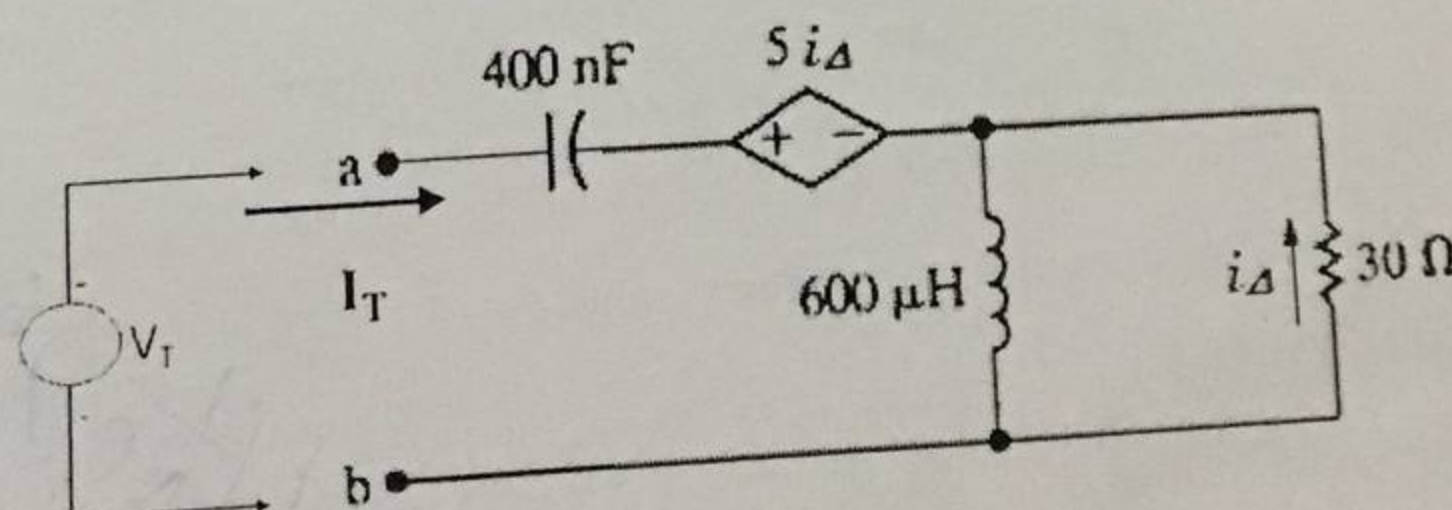


$i(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t}$ (overdamped), $i(t) = B_1 e^{-\alpha t} \cos \omega_d t + B_2 e^{-\alpha t} \sin \omega_d t$ (underdamped), $i(t) = D_1 t e^{-\alpha t} + D_2 e^{-\alpha t}$ (critically damped).	$v_C = V_f + A'_1 e^{s_1 t} + A'_2 e^{s_2 t}$ (overdamped), $v_C = V_f + B'_1 e^{-\alpha t} \cos \omega_d t + B'_2 e^{-\alpha t} \sin \omega_d t$ (underdamped), $v_C = V_f + D'_1 t e^{-\alpha t} + D'_2 e^{-\alpha t}$ (critically damped).
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- (a) Find $i(0+)$, $v_C(0+)$, $\frac{di(0+)}{dt}$ and $\frac{dv_C(0+)}{dt}$.
- (b) Write the second order differential equation for $i(t)$ in time domain for $t \geq 0$ and find the characteristic equation as well.
- (c) Give the roots of the characteristic equation and identify the type of response from the roots.
- (d) Give the complete expression for $i(t)$ and $v_C(t)$ for $t \geq 0$.
- (e) Identify the transient and steady-state part of $i(t)$ and $v_C(t)$.

Question # 2 (a) [10 marks]

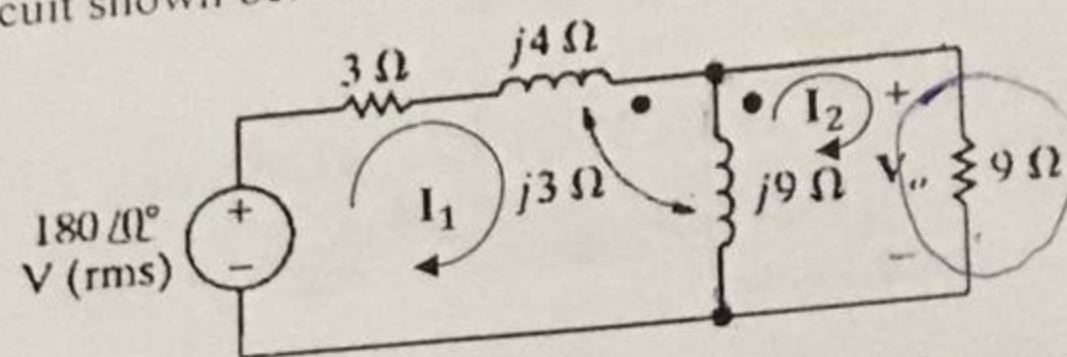
Find Z_{ab} , using test source method in the circuit shown below. The circuit is operating at a frequency of $\omega = 100 \text{ krad/s}$. Hint: Assume sinusoidal voltages and currents, and convert the circuit to phasor domain first.



$$-\alpha = -\frac{1}{2RC}$$

Question # 2 (b) [10 marks]

For the frequency-domain circuit shown below, find the following:



- Using mesh analysis, write the two mesh equations for I_1 and I_2 .
- Find $V_o(rms)$.
- Find the average power dissipated in the 9Ω resistor.

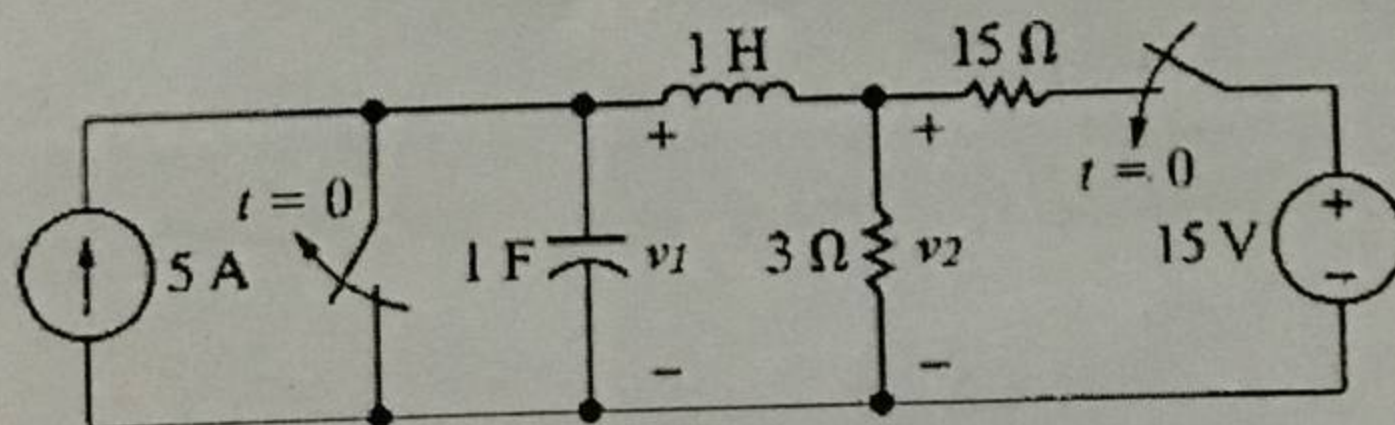
Question # 3 [8+4+8 = 20 marks]

A Y- Δ connected balanced three phase system is characterized as follows:

- Internal impedance of the generator = $0 + j0 \Omega/\phi$;
 - Source voltage in the c-phase is $20\angle -90^\circ$ V;
 - Source phase sequence is abc;
 - Line impedance is $1 + j3 \Omega/\phi$;
 - Load impedance is $117 - j99 \Omega/\phi$.
- Draw the single phase equivalent for the a-phase.
 - Calculate the a-phase line current.
 - Calculate the a-phase line voltage at the load.

Question # 4 [10+6+2+2 = 20 marks]

The dc current and voltage sources are applied simultaneously to the circuit shown at $t = 0$. No energy is stored in the circuit at the instant of application.



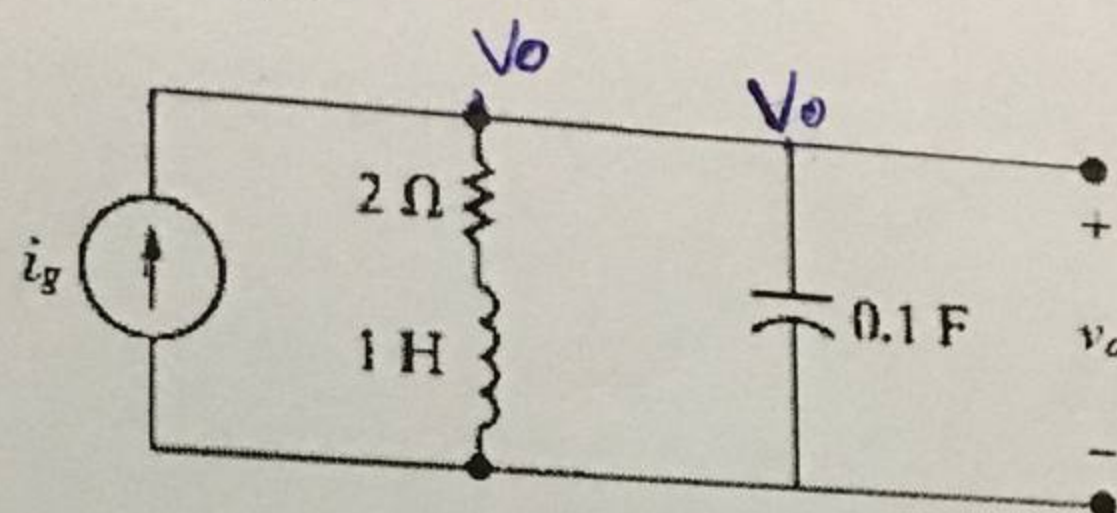
- Using NODE analysis, find the s-domain expressions for V_1 and V_2 .
- For $t > 0$, find the time-domain expressions for v_1 and v_2 .
- Calculate $v_1(0+)$ and $v_2(0+)$.
- Compute the steady-state values of v_1 and v_2 .

Handwritten notes: $s^2 + \frac{1}{RC}s + \frac{1}{LC} = 0$

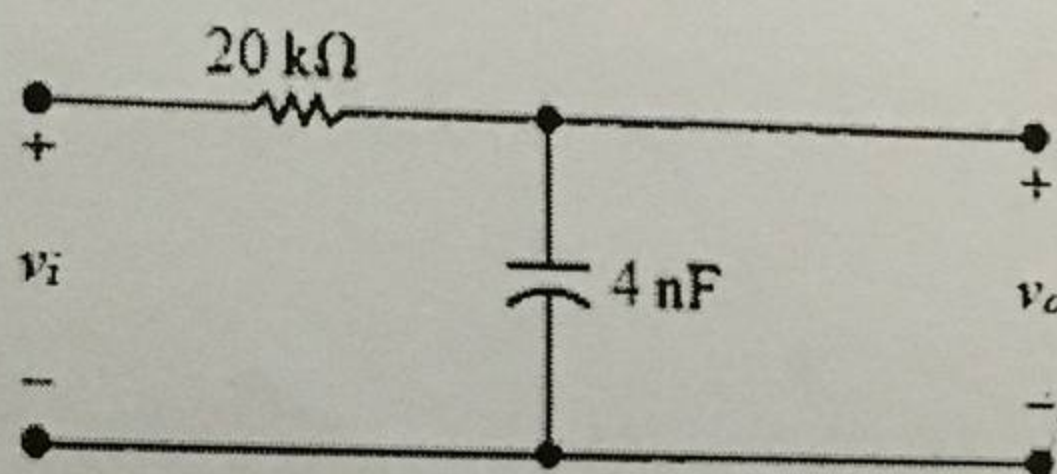
Question # 5 [10+10 = 20 marks]

- (a) Find the transfer function, $H(s)$, and the unit-impulse response, $h(t)$, for the following circuit.

Hint: $h(t) = \mathcal{L}^{-1}[H(s)]$ and $H(s) = \frac{V_o(s)}{I_g(s)}$



- (b) For the filter circuit shown below, answer the following:



- (i) Is it a high-pass or a low-pass filter?
- (ii) Find the cut-off frequency (in hertz).
- (iii) Find $H(s)$ and calculate $H(j\omega)$ at ω_c , $0.2\omega_c$, and $8\omega_c$.
- (iv) If $v_i = 0.48 \cos \omega t$ V, write the steady-state expression for v_o when $\omega = 8\omega_c$.

Hint: $v_{o_{ss}}(t) = A|H(j\omega)| \cos[\omega t + \theta(\omega)]$